Week 3 – OFDM Synchronization

1 OFDM Sync Long

The `OFDM Sync Long` block performs frequency offset correction and symbol alignment.

1.1 Frequency Offset Correction

Since the transmitter and receiver oscillators may operate at slightly different frequencies, the received signal experiences phase rotation. The algorithm estimates this offset using the periodicity of the Short Training Field (STF), which repeats every 16 samples.

The frequency offset is estimated using:

$$P = \sum_{n=0}^{N_{\text{short}}-17} s[n]s^*[n+16]$$

$$\Delta f = \frac{1}{16}\arg(P)$$

The correction is then applied to each sample:

$$s[n] \leftarrow s[n]e^{jn\Delta f}$$

1.2 Implementation and Parameters

Frequency offset correction is implemented inside the `OFDM Sync Long` block, using the STF. Symbol alignment uses the Long Training Field (LTF) through matched filtering.

The main parameters that can be varied are the number of samples used for estimation (N_{short}), the correlation separation (16 samples), and the delay block value (240 samples).

1.3 Delayed Input

The delay provides a time-shifted signal to the synchronization block. Changing its value affects the relative timing of the inputs and may cause synchronization and symbol alignment errors.